

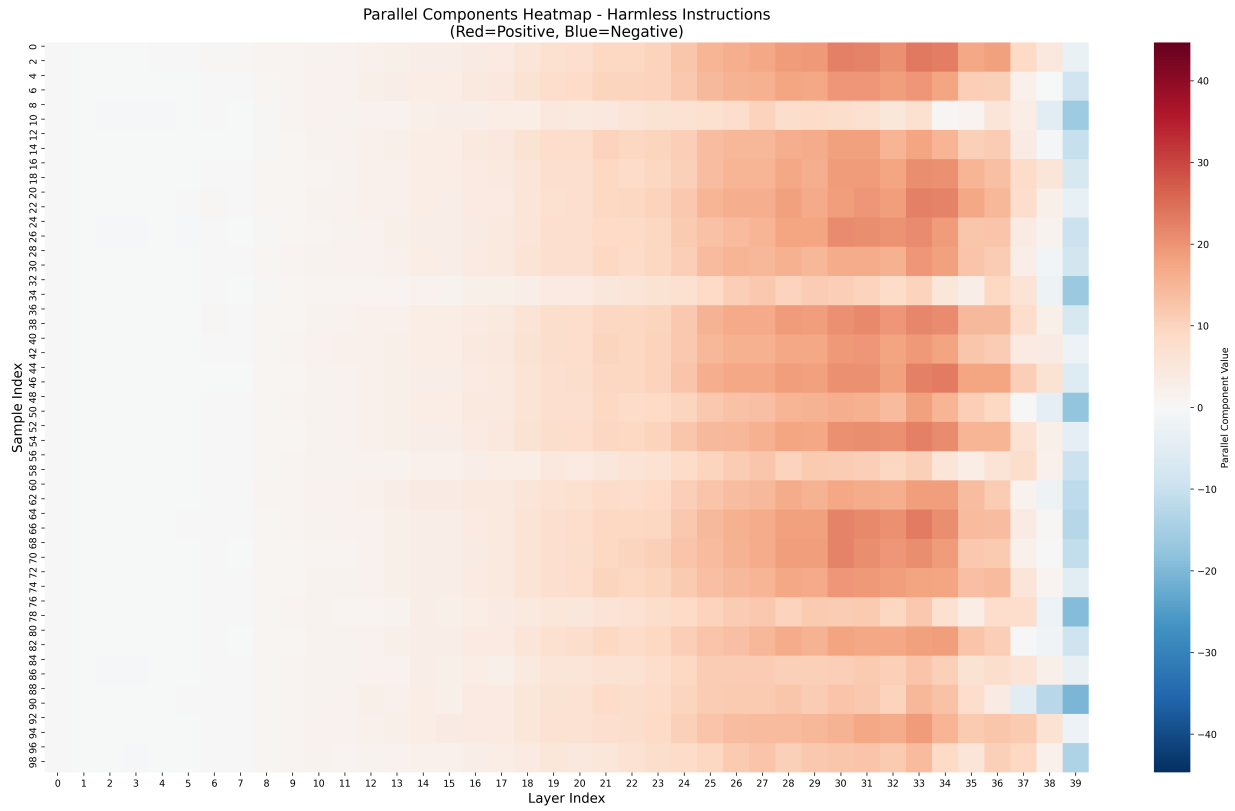


Figure 12. Refusal components for harmless instructions (Qwen3-14B). Harmless instructions shift from positive values in middle layers (safety checking) to negative values in final layers (safety verification allowing compliance).

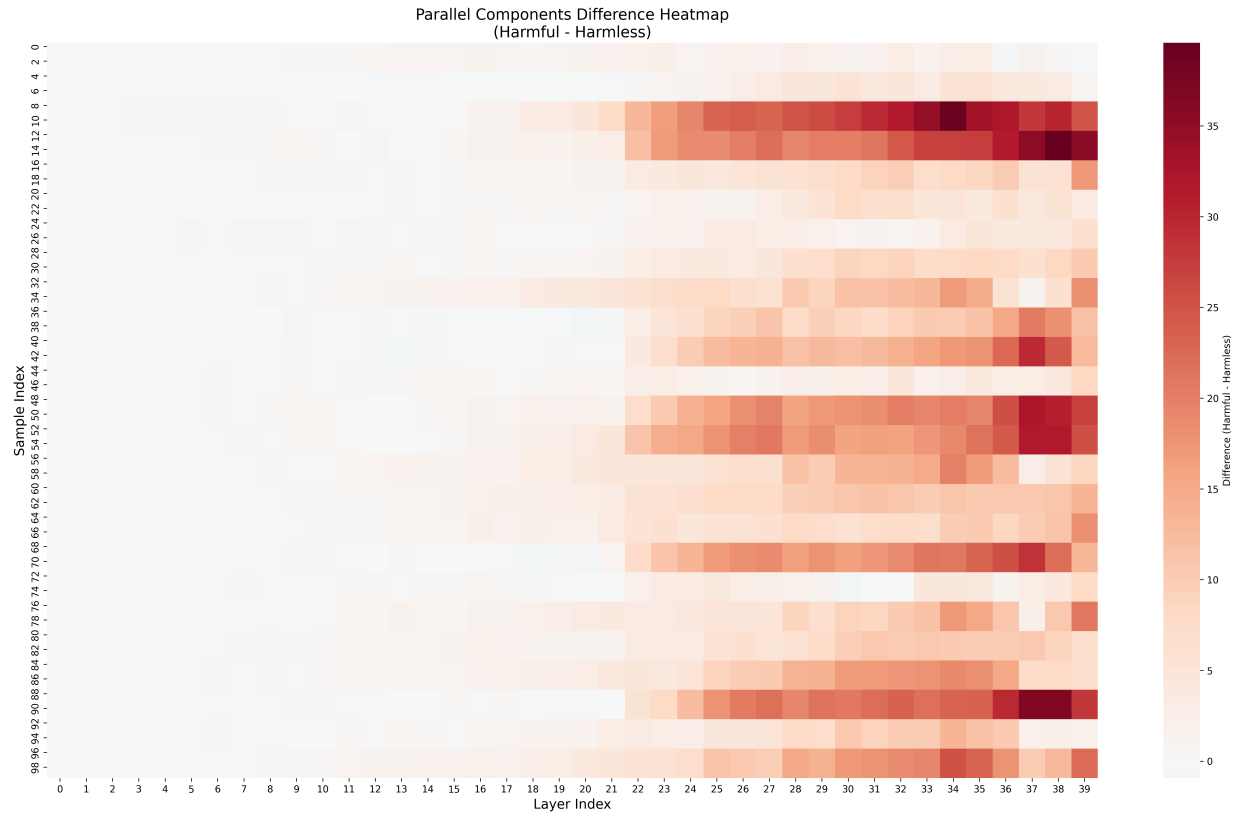


Figure 13. Difference heatmap (harmful–harmless). Red regions show where harmful instructions elicit stronger refusal components than harmless ones, especially in layers 25–35.

F. Refusal Component Experiments Based on Jailbreak Success

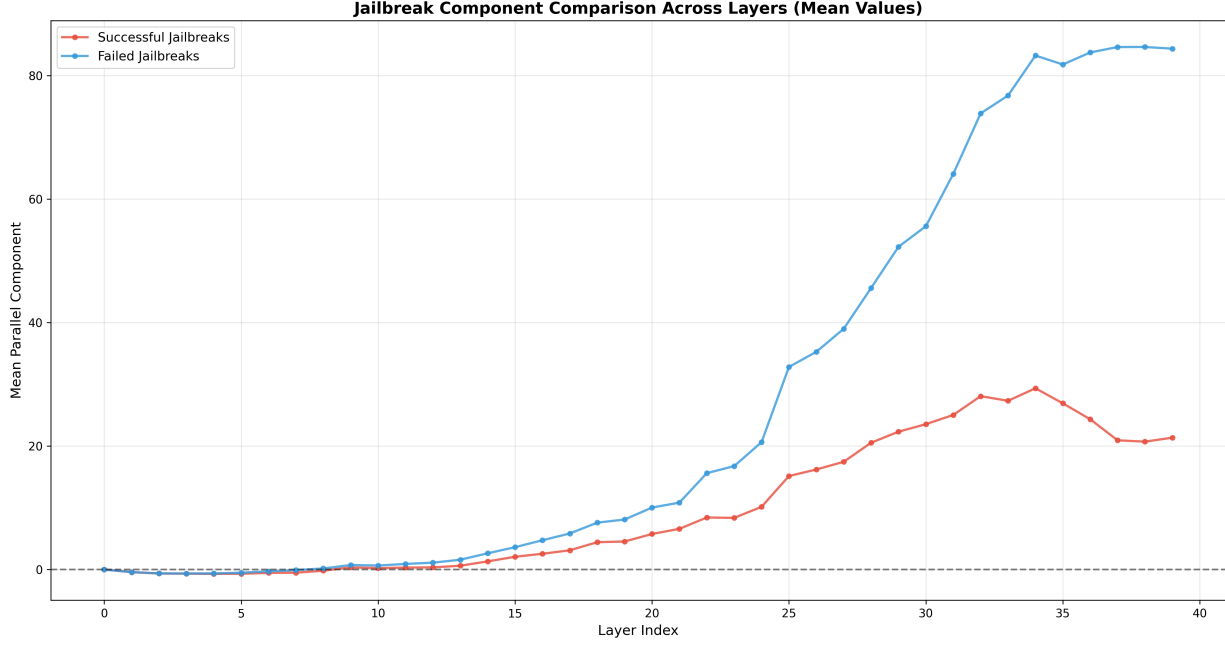


Figure 14. **Refusal component by outcome.** Mean parallel components across layers for successful vs. failed jailbreak attempts. The activation is measured at the last token of the full generation (prompt + CoT + final response).

G. Additional Results on CoT Length

G.1. Algorithm for Template Generation

Algorithm 1 Generate templates and compute refusal components for different CoT lengths

Require: p : puzzle question; i : harmless/harmful/stealthy harmful instruction

Ensure: $\{R(T^{(L)}) : L \in \mathcal{L}\}$ {refusal components}

- 1: $\mathcal{L} = \{1k, 3k, 11k, 21k, 31k, 47k\}$
 - 2: $P_0 = p \oplus i$ { $\oplus$ denotes concatenation}
 - 3: $(C_p^{(\text{full})}, C_i, r) = \text{LRM}(P_0)$ { $C_p^{(\text{full})}$: full puzzle-solving CoT ($L_{\max} = 47k$); C_i : instruction-analysis CoT; r : final response}
 - 4: $T^{(L_{\max})} = P_0 \oplus C_p^{(\text{full})}$ {template of length 47k}
 - 5: Define $\text{trim_mid}(\cdot; L)$ {remove middle tokens to yield target length L }
 - 6: **for** each $L \in \mathcal{L}$ **do**
 - 7: **if** $L = L_{\max}$ **then**
 - 8: $C_p^{(L)} = C_p^{(\text{full})}$
 - 9: **else**
 - 10: $C_p^{(L)} = \text{trim_mid}(C_p^{(\text{full})}; L)$
 - 11: **end if**
 - 12: $T^{(L)} = P_0 \oplus C_p^{(L)}$
 - 13: $h_{\text{last}}^{(L)} = \text{LRM}_{\text{forward}}(T^{(L)})$ {activation of last token}
 - 14: $R(T^{(L)}) = \langle h_{\text{last}}^{(L)}, v_{\text{refusal}} \rangle$
 - 15: Record $R(T^{(L)})$
 - 16: **end for**
 - 17: **return** $\{R(T^{(L)}) : L \in \mathcal{L}\}$
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